

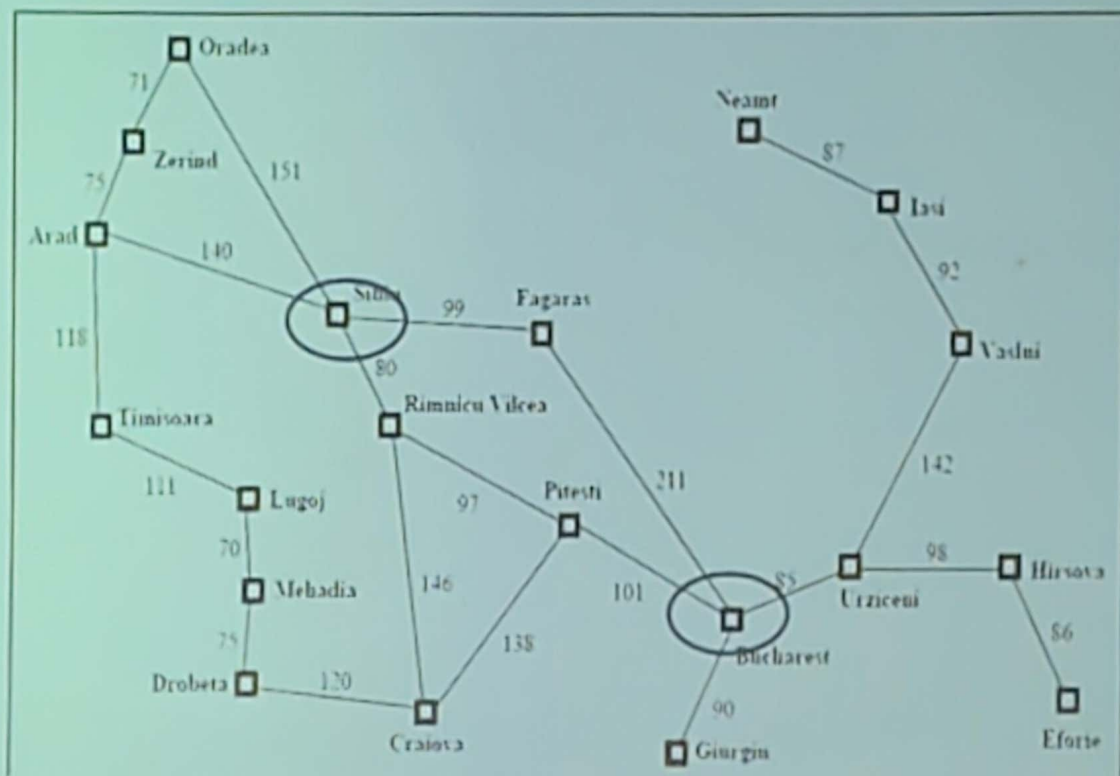
A brief intro

A pen is being sold in a shop for 100 rupees

- What makes you decide to buy this pen ?
- Would it be the same value for every one of you ?
- On what factors does it depends on ?
- How should you capture it for an automated agent?

Techniques to take decisions in the face of **uncertainty**:

A Map of Romania



Which route to take ?

- Route 1: Sibiu → Fagaras → Bucharest : 310
- Route 2: Sibiu → Rimnicu Vilcea → Pitesti → Bucharest : 278
- Ideally would take the shortest route
- What if traffic is a factor now ?
- Suppose you have accurate information that
 - Route 1 has fast moving traffic while route 2 has slow moving traffic
- A logical agent may reason
 - *Slow(Route 2), fast(Route 1)*
 - *Slow(x) => Avoid (x)*
 - *Avoid(x) ^ fast(y) => select(y)*
 - *Agent selects Route 1*
- Do you think this is a good way to reason?
- Why or why not?

Decision Theory

(How to make decisions)

Decision Theory

= Probability theory + Utility Theory
(deals with chance) *(deals with outcomes)*

- *Fundamental idea:*
 - *The MEU (Maximum expected utility) principle*
 - *Agent is rational if and only if it chooses the action that yields the highest expected utility, averaged over all possible outcomes of the action*
 - *Weigh the utility of each outcome by the probability that it occurs*

Uncertainty

- May not have categorical answers
 - *May not know exactly which route is slow*
 - *50% chance that one route is slow*
 - *Even if we know it is slow, how slow ?*
- **Uncertainty** changes the way an agent makes a decision
- Probability theory used in helping agents reason with uncertainty
 - *Summarize uncertainty due to our ignorance or laziness*
 - *May be very very costly to remove such uncertainty*
- **Rational decision** depends on
 - Relative importance of various goals
 - Likelihood or degree to which they will be achieved

Utility Theory – von Neumann (Game Theory)

Let X be the set of outcomes. $\succ$ be the preference of a player over the set of outcomes

Axioms

- **Completeness:** every pair of outcomes is ranked
- **Transitivity:** If $x_1 \succ x_2$ and $x_2 \succ x_3$ then $x_1 \succ x_3$
- **Substitutability:** If $x_1 \sim x_2$ then any lottery in which x_1 is substituted by x_2 is equally preferred
- **Decomposability:** two different lotteries assign same probability to each outcome, then player is indifferent between these two lotteries
- **Monotonicity:** If $x_1 \succ x_2$ and $p > q$ then $[x_1 : p, x_2 : 1 - p] \succ [x_1 : q, x_2 : 1 - q]$
- **Continuity:** If $x_1 \succ x_2 \succ x_3$, $\exists p \ni x_2 \sim [x_1 : p, x_3 : 1 - p]$

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Von Neumann and Morgenstern

Theorem

Given a set of outcomes X and a preference relation on X that satisfies above six axioms, there exists a utility function $u : X \rightarrow [0, 1]$ with the following properties:

- 1 $u(x_1) \geq u(x_2)$ iff $x_1 \succsim x_2$
- 2 $U([x_1 : p_1; x_2 : p_2; \dots; x_m : p_m]) = \sum_{j=1}^m p_j u(x_j)$

Conditional Probability

- What if agent has some evidence?
- What if agent knows that if the last 2 days are sunny there is higher probability that it will be sunny today ?
- Posterior or conditional probability
- $P(A|B)$ probability of A given all we know is B
 - $P(X=\text{sunny} | \text{Last 2 days were sunny})$
- If we know B and also know C, then $P(A | B \wedge C)$

Diagnosis domain: Introducing Cavity Variable

Doctor diagnosing a dental patient's toothache

Propositional logic rules

Toothache $\rightarrow$ Cavity [Toothache implies Cavity] (Incomplete rule)

Toothache $\rightarrow$ Cavity $\vee$ Gum Problem $\vee$ Abscess ... (Lots of such problems)

Turning the rule around, Cavity $\rightarrow$ Toothache

Joint Probability Distribution

- Notation for distributions on multiple variables
 - $P(\text{Weather}, \text{Cavity})$ i.e. $P(\text{Weather} \wedge \text{Cavity})$
- Joint probability distribution is a table
 - *Assigns probabilities to all possible assignment of values for combinations of variables*
 - *Weather has values {sunny, rain, cloudy, snow}*
 - *Cavity has values {true, false}*
 - *4 * 2 table of probabilities called joint probability distribution*
- $P(X_1, X_2, \dots, X_n)$ assigns probabilities to all possible assignment of values to variables $X_1, X_2, \dots, X_n$

Inference using full joint distribution

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	.108	.012	.072	.008
$\neg$ <i>cavity</i>	.016	.064	.144	.576

- 3 variables : {Cavity, Toothache, Catch}
- Dentist probe catches or not
- $2 * 2 * 2$ table

Marginalization

- Denominator can be viewed as a normalization constant $1/P(\text{toothache})$
- $P(\text{Cavity} | \text{Toothache}) = \alpha P(\text{Cavity}, \text{toothache}) = \alpha [P(\text{Cavity}, \text{toothache}, \text{catch}) + P(\text{Cavity}, \text{toothache}, \neg \text{catch})] = .108 + .012 = \alpha 0.12$
- $\alpha P(\neg \text{Cavity}, \text{toothache}) = \alpha [.016 + .064] = \alpha 0.08$
- Normalizing $\alpha \langle .12, .08 \rangle$ gives $\langle .6, .4 \rangle$
- **Issue:** For a domain with n Boolean variables, input table size of $O(2^n)$
- Full joint distribution in tabular form not a practical tool for building reasoning systems

Independence

- Helps in reducing size of domain representation and complexity of inference problem
- Let's add weather into the example : sunny, rain, cloudy, snow $\rightarrow 2 * 2 * 2 * 4$ entries in table
- $P(\text{toothache, catch, cavity, cloudy}) = P(\text{cloudy} \mid \text{toothache, catch, cavity}) P(\text{toothache, catch, cavity})$
- $P(\text{cloudy} \mid \text{toothache, catch, cavity}) = P(\text{cloudy})$
- 32 element table becomes 8 element + 4 element table

Bayes' Rule

- Given that
 - $P(A \wedge B) = P(A|B) * P(B)$
 - $P(A \wedge B) = P(B|A) * P(A)$
 - $P(B|A) = (P(A|B) * P(B)) / P(A)$
- We can determine $P(B|A)$ given $P(A|B)$, $P(B)$ and $P(A)$
- $P(\text{effect} | \text{cause})$ may be **causal knowledge**, $P(\text{cause} | \text{effect})$ **diagnostic knowledge**
- Often have conditional probabilities on causal relationships $\rightarrow P(\text{symptoms} | \text{disease})$
 - Need to infer diagnostic knowledge many times

More General Forms of Bayes' Rule

- $P(Y | X) = P(X | Y) * P(Y) / P(X)$
- Bayes' rule for multi-valued variables
- Generalize to some background evidence e
- $P(Y | X, e) = P(X | Y, e) * P(Y | e) / P(X | e)$

Conditional Independence

- Toothache and Catch are independent given the presence of Cavity
 - Each is directly caused by cavity, toothache is caused by cavity and a dentist catches cavity using probe
 - A toothache cannot be caught by a probe nor a probe results in a toothache
- Conditional independence captured as : $P(\text{toothache} \wedge \text{catch} \mid \text{cavity}) = P(\text{toothache} \mid \text{catch}, \text{cavity}) P(\text{catch} \mid \text{cavity})$
- $P(\text{toothache} \mid \text{catch}, \text{cavity}) = P(A \mid B, C)$
 - If A and C are conditionally independent given B
 - Then, probability of A is not dependent on C
 - i.e. $P(A \mid B \wedge C) = P(A \mid B)$
- Therefore $P(\text{toothache}, \text{catch} \mid \text{cavity}) = P(\text{toothache} \mid \text{cavity}) P(\text{catch} \mid \text{cavity})$

Combining Evidence

- Revisiting the Meningitis example --
 - S: Proposition that patient has stiff neck
 - H: Proposition that patient has severe headache
 - M: Proposition that patient has meningitis
 - Meningitis causes stiff-neck, 50% of the time
 - Meningitis causes head-ache, 70% of the time
 - Probability of Meningitis should go up, if both symptoms reported -- how to combine such symptoms?

Application of MEU Principle

- $$\begin{aligned} EU(\text{Plan1}) &= P(\text{home-early} \mid \text{plan1}) * U(\text{home-early}) \\ &\quad + P(\text{stuck1} \mid \text{plan1}) * U(\text{stuck1}) \\ &= 0.8 * 100 + 0.2 * -1000 = -120 \end{aligned}$$
- $$\begin{aligned} EU(\text{Plan2}) &= P(\text{home-somewhat-early} \mid \text{plan2}) * U(\text{home-somewhat-early}) \\ &\quad + P(\text{stuck2} \mid \text{plan2}) * U(\text{stuck2}) \\ &= 0.7 * 50 + 0.3 * -10 = 32 \end{aligned}$$

Revisiting Romania example

- If plan1 and plan2 are the two plans:

- Plan 1 uses route 1

- $P(\text{home-early}|\text{plan1}) = .8$, while $P(\text{stuck1}|\text{plan1}) = .2$
- Route 1 will be quick if flowing, but stuck for 1 hour if slow
- $U(\text{home-early}) = 100$, $U(\text{stuck1}) = -1000$
- Assigned numerical values to outcomes!

$$\begin{aligned} EU &= -100 \\ 100 \times 0.8 \\ - 1000 \times 0.2 \end{aligned}$$

- Plan 2 uses route 2

- $P(\text{home-somewhat-early}|\text{plan2}) = .7$, $P(\text{stuck2}|\text{plan2}) = .3$
- Route 2 will be somewhat quick if flowing, but not bad even if slow
- $U(\text{home-somewhat-early}) = 50$, $U(\text{stuck2}) = -10$

$$\begin{aligned} EU &= 50 \times 0.7 \\ &\quad - 10 \times 0.3 \\ &= 32 \end{aligned}$$

Application of MEU Principle

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Risk Aversion

- We are **risk averse**
- Our utility functions for money are as follows (!!):
 - Our first million means a lot $U(\$1M) = 10$
 - Second million not so much $U(\$2M) = 15$ (NOT 20)
 - Third million even less so $U(\$3M) = 18$ (NOT 30)
 - ...
- Additional money is not buying us as much utility
- If we plot amount of money on the x-axis and utility on the y-axis, we get a concave curve

More Risk Aversion

- Key: Slope of utility function is **continuously decreasing**
 - We will refuse to play a monetarily fair bet
- Suppose we start with x dollars
 - We are offered a game:
 - 0.5 chance to win 1000 dollars ($c = 1000$)
 - 0.5 chance to lose 1000 dollars ($c = 1000$)
 - Expected monetary gain or loss is zero (hence monetarily fair)
- Should be neutral to it, but seems we are not! Why?
 - $U(x + c) - U(x) < U(x) - U(x - c)$
 - $U(x + c) + U(x - c) < 2 U(x)$
 - $[U(x + c) + U(x - c) / 2] < U(x)$
 - $EU(\text{playing the game}) < EU(\text{not playing the game})$

Example: What SEQUENCE of actions should our agent take?

- Agent can take action N, E, S, W
- Each action costs $-1/25$

